Project 1: Convolution

Part 1 – High-Level Implementation (Java)

```java
/**
 * Demonstrating 1-D convolution on an array of integers
 */
public class Convolution {
    /**
     * Performs simple convolution on integer array.
     * New array replaced with average of itself and neighbors.
     *
     * @param A input array of int to use
     * @return new array containing the average values
     */
    public static int[] convolution(int[] A) {

        int N = A.length;
        int[] B = new int[N];

        for (int i = 0; i < N; i++) {
            int sum = A[i];

            if (i > 0) {
                sum += A[i - 1]; // adding left nieghbor
            }

            if (i < N - 1) {
                sum += A[i + 1]; // adding right neigbor
            }

            B[i] = sum / 3; // average to be added to B
        }

        return B;
    }

    public static void printArray(int[] arr) {
        for (int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
        }

        System.out.println();
    }
```

```
    public static void main(String[] arts) {

        // Test_1
        int[] A1 = { 9, 20, 21, 8, 11, 12, 3, 15 };
        int[] B1 = convolution(A1);

        System.out.println("Test_1");
        System.out.print("Array A: ");
        printArray(A1);

        System.out.print("Array B: ");
        printArray(B1);
        System.out.println();

        // Test_2
        int[] A2 = { 1, 1, 1, 1, 1 };
        int[] B2 = convolution(A2);

        System.out.println("Test_2");
        System.out.print("Array A: ");
        printArray(A2);

        System.out.print("Array B: ");
        printArray(B2);
        System.out.println();

        // Test_3
        int[] A3 = { 100, 50, 25, 130, 200, 10 };
        int[] B3 = convolution(A3);

        System.out.println("Test_3");
        System.out.print("Array A: ");
        printArray(A3);

        System.out.print("Array B: ");
        printArray(B3);

    }
}
```

Part 2 – RISCV Assembly Implementation

```
.data



arrA: .string "Array A: "

arrB: .string "\nArray B: "

space: .string " "

A: .word 100, 50, 25, 130, 200, 10    # Change for testing

# Output change for testing

B: .space 24    # 5 int * 4 bytes

N: .word 6        # array length



.text

.globl main



main:

   la s0, A
```

```
    la s1, B

    lw s2, N

    li t0, 0          # Initialize loop i starting at 0


loop:


    bge t0, s2, end_loop          # if i >= N end


    slli t1, t0, 2  # byte offset for array access

    add t2, s0, t1          # Add offeset to A = A[i]

    lw t3, 0(t2)


    beq t0, x0, skip_left          # if i == 0 skip, left neighbor check


    addi t4, t0, -1

    slli t4, t4, 2              # offset

    add t5, s0, t4          # Memory address of A[i - 1]
```

```
    lw t6, 0(t5)

    add t3, t3, t6          # Add left neighbor

skip_left:

    addi t4, s2, -1         # N - 1 check for last input

    beq t0, t4, skip_right          # if i == N - 1 skip, right neighbor check

    addi t4, t0, 1          # index i + 1

    slli t4, t4, 2  # offset

    add t5, s0, t4          # Memory address A[i + 1]

    lw t6, 0(t5)

    add t3, t3, t6          # Add right neighbor
```

```
skip_right:

    li t4, 3

    div t3, t3, t4

    add t5, s1, t1          # Memory address B[i]

    sw t3, 0(t5)  # Store into arr B

    addi t0, t0, 1          # increment index looping

    j loop

end_loop:

    la a0, arrA             # Array A for print

    li a7, 4                # System call print string

    ecall

    li t0, 0          # Reset counter print Array A
```

```
printA:

    bge t0, s2, printB    # move to Array B

    slli t1, t0, 2  # offset A[i]

    add t2, s0, t1

    lw a0, 0(t2)

    li a7, 1          # System call print int

    ecall

    la a0, space

    li a7, 4

    ecall

    addi t0, t0, 1
```

```
    j printA

printB:

    la a0, arrB
    li a7, 4
    ecall

    li t0, 0

printBloop:

    bge t0, s2, exit        # if all print exit

    slli t1, t0, 2
    add t2, s1, t1
    lw a0, 0(t2)
```

```
    li a7, 1
    ecall

    la a0, space
    li a7, 4
    ecall

    addi t0, t0, 1
    j printBloop

exit:

    li a7, 10
    ecall
```

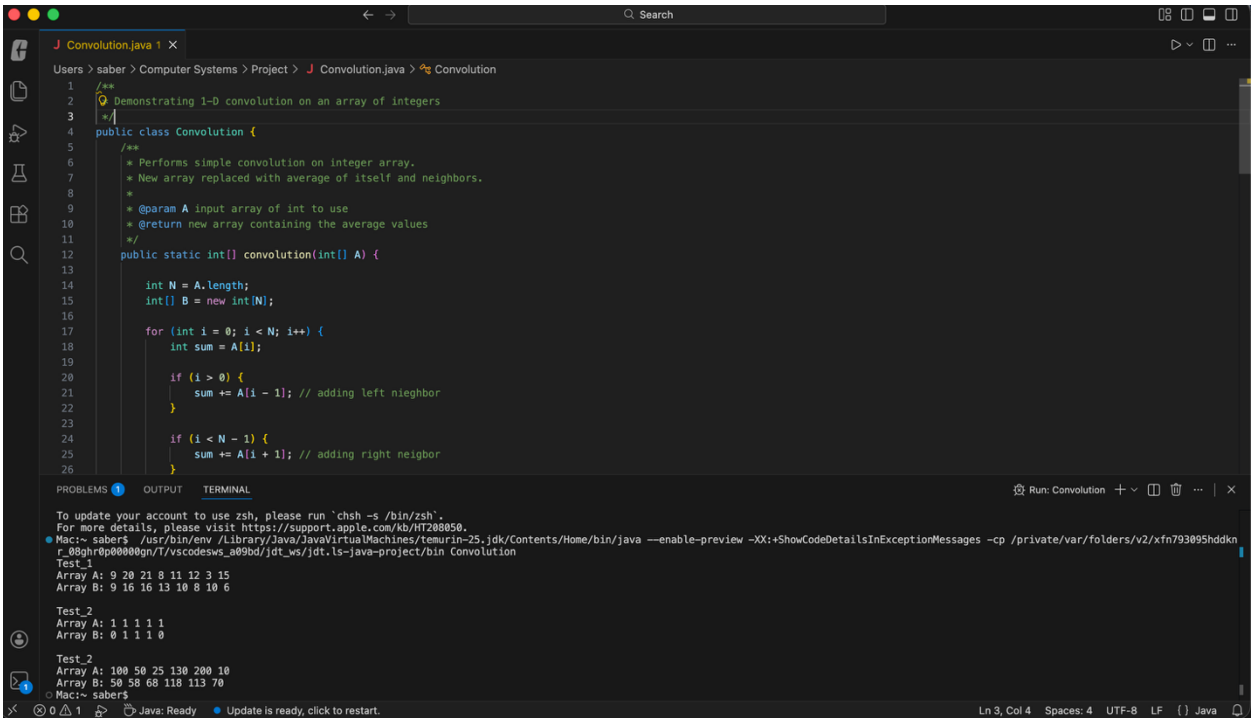
Part 3 – Write Up

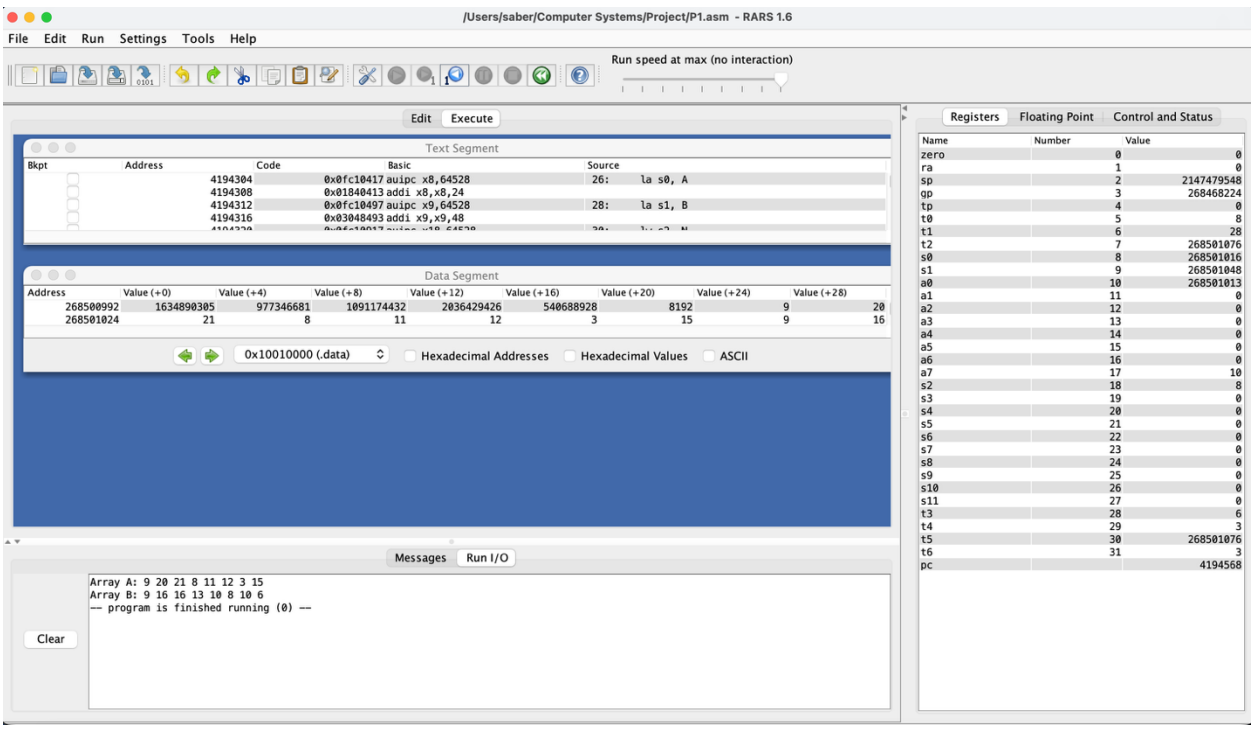
The algorithm for convolution project is the input array's elements and its neighbors being averaged. Each new output is then added to the new array B by taking the current index of array A and adding the current element with its neighboring elements if it exists. The total sum will then be divided by three and produce an average. Then certain conditions must be checked for the first and last elements as they will not have both neighbors. The conditional checks will be making sure the indices checked do not try to perform out of range. 'beq t0, x0, skip_left' t0 being [i] to make sure it does not access an index like A[-1] for the first index. As for the last element check there is total N length and [i] current index if i == N -1 then the process will stop. 'addi t4, s2 -1' and 'beq t0, t4, skip_right'

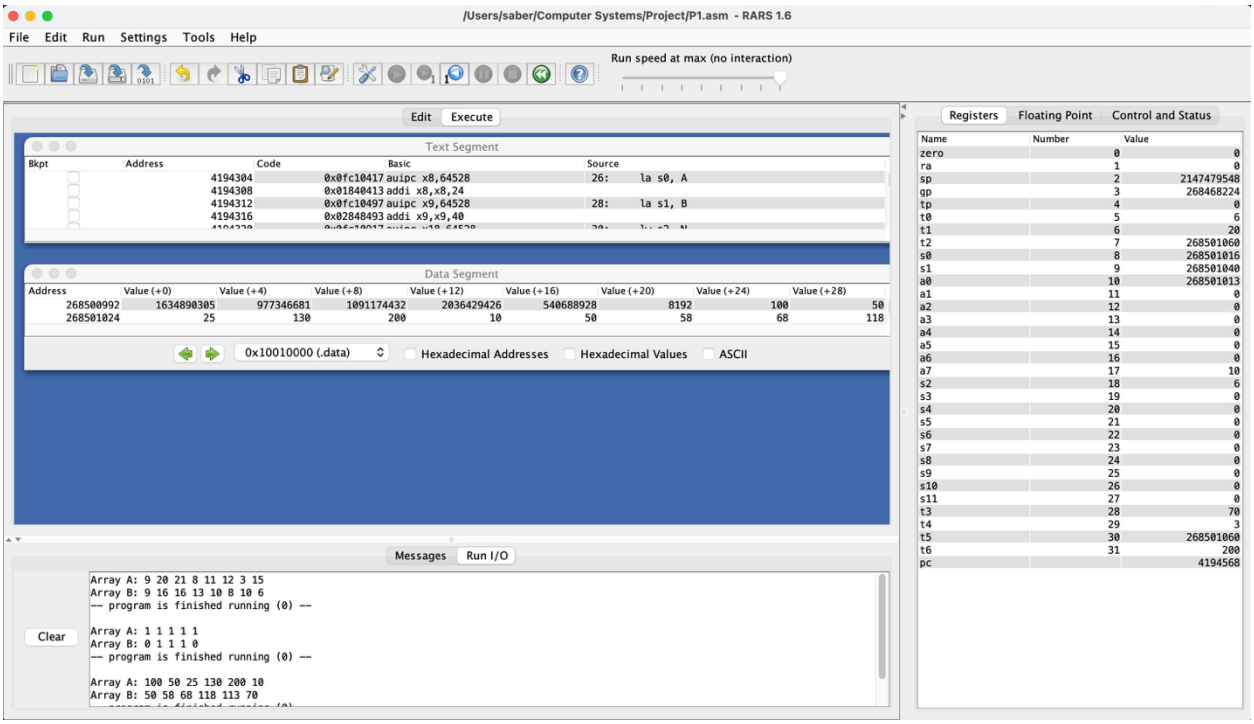
The high-level implementation was simpler. Jave being used has auto management of array indexing and arithmetic. These processes made it less troublesome dealing with the RISCV Assembly of managing the memory addresses. As in accessing A[i], A [i - 1]. and A[i + 1]. Translating the algorithm to RISC-V required computing every step and off-set. Manual calculations of these steps can come with errors as there are more steps involved. A challenge at first was figuring out the first and last elements looping requirements to skip calculations if it did not have a second neighbor.

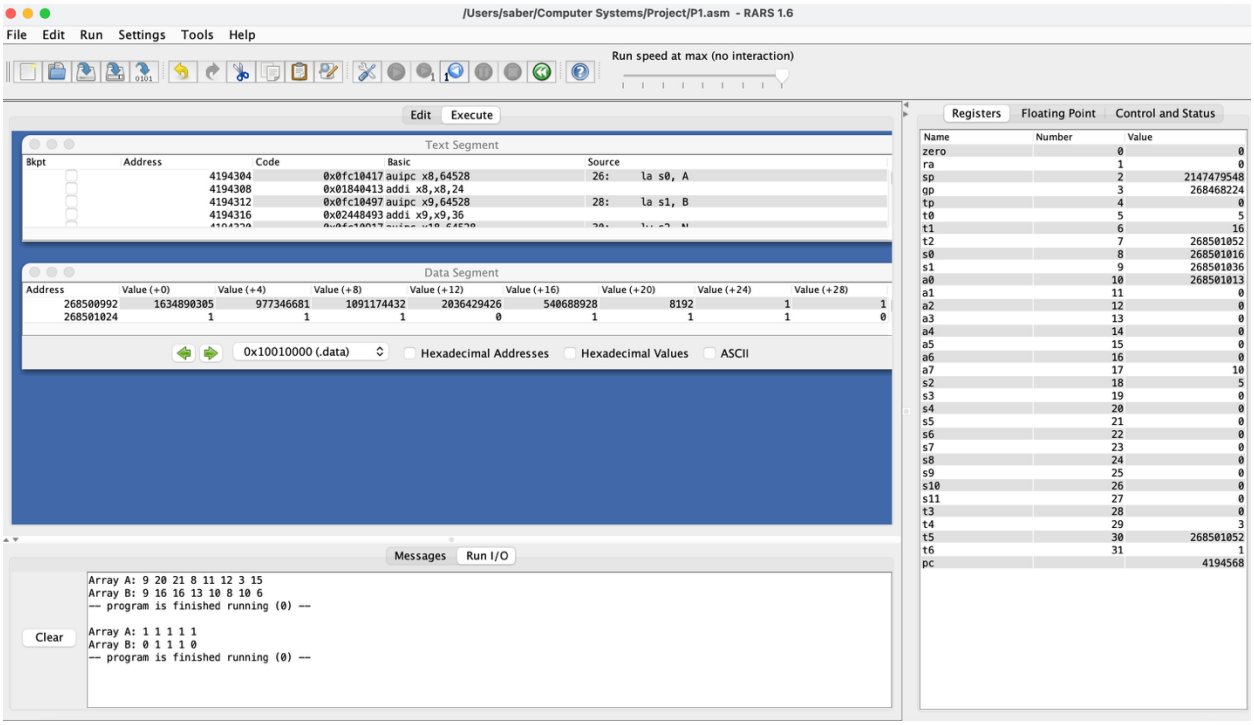
Screenshots of Tests

*First screenshot has typo in the last test should be Test_3.






/Users/saber/Computer Systems/Project/P1.asm - RARS 1.6
File Edit Run Settings Tools Help
Run speed at max (no interaction)
Edit Execute
Text Segment
Bkpt Address Code Basic Source
4194304 0x0fc10417 auipc x8,64528 26: la s0, A
4194308 0x01840413 addi x8,x8,24
4194312 0x0fc10497 auipc x9,64528 28: la s1, B
4194316 0x02848493 addi x9,x9,40
Data Segment
Address Value (+0) Value (+4) Value (+8) Value (+12) Value (+16) Value (+20) Value (+24) Value (+28)
268500992 1634890305 977346681 1091174432 2036429426 540688928 8192 100 50
268501024 25 130 200 10 50 58 68 118
0x10010000 (.data) Hexadecimal Addresses Hexadecimal Values ASCII
Messages Run I/O
Clear
Array A: 9 20 21 8 11 12 3 15
Array B: 9 16 16 13 10 8 10 6
-- program is finished running (0) --
Array A: 1 1 1 1 1
Array B: 0 1 1 1 0
-- program is finished running (0) --
Array A: 100 50 25 130 200 10
Array B: 50 58 68 118 113 70
Registers Floating Point Control and Status
Name Number Value
zero 0 0
ra 1 0
sp 2 2147479548
gp 3 268468224
tp 4 0
t0 5 6
t1 6 20
t2 7 268501060
s0 8 268501016
s1 9 268501040
a0 10 268501013
a1 11 0
a2 12 0
a3 13 0
a4 14 0
a5 15 0
a6 16 0
a7 17 10
s2 18 6
s3 19 0
s4 20 0
s5 21 0
s6 22 0
s7 23 0
s8 24 0
s9 25 0
s10 26 0
s11 27 0
t3 28 70
t4 29 3
t5 30 268501060
t6 31 200
pc 4194568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4194316 0x02848493 addi x9,x9,36
Data Segment
Address Value (+0) Value (+4) Value (+8) Value (+12) Value (+16) Value (+20) Value (+24) Value (+28)
268500992 1634890305 977346681 1091174432 2036429426 540688928 8192 1 1
268501024 1 1 1 0 1 1 1 0
0x10010000 (.data) Hexadecimal Addresses Hexadecimal Values ASCII
Messages Run I/O
Clear
Array A: 9 20 21 8 11 12 3 15
Array B: 9 16 16 13 10 8 10 6
-- program is finished running (0) --
Array A: 1 1 1 1 1
Array B: 0 1 1 1 0
-- program is finished running (0) --
Registers Floating Point Control and Status
Name Number Value
zero 0 0
ra 1 0
sp 2 2147479548
gp 3 268468224
tp 4 0
t0 5 5
t1 6 16
t2 7 268501052
s0 8 268501016
s1 9 268501036
a0 10 268501013
a1 11 0
a2 12 0
a3 13 0
a4 14 0
a5 15 0
a6 16 0
a7 17 10
s2 18 5
s3 19 0
s4 20 0
s5 21 0
s6 22 0
s7 23 0
s8 24 0
s9 25 0
s10 26 0
s11 27 0
t3 28 0
t4 29 3
t5 30 268501052
t6 31 1
pc 4194568